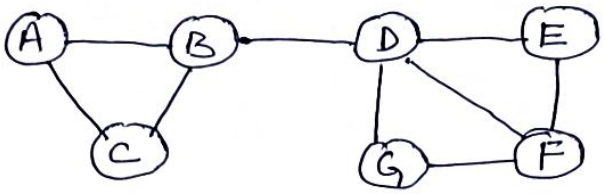
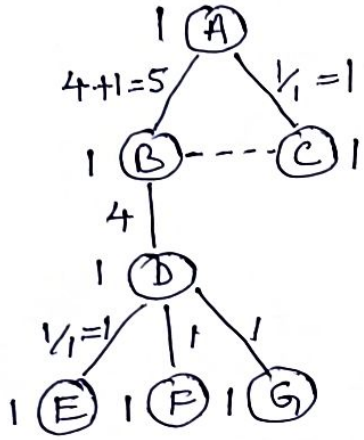
# Girvan Newman Algorithm – Example.

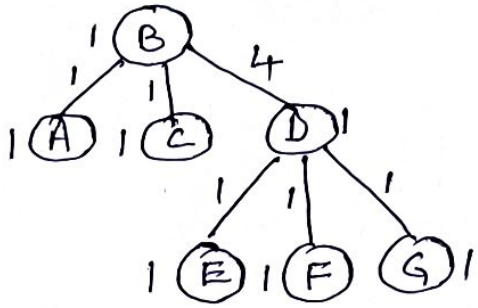
Find communities in the graph using Girvan Newman Alg :-



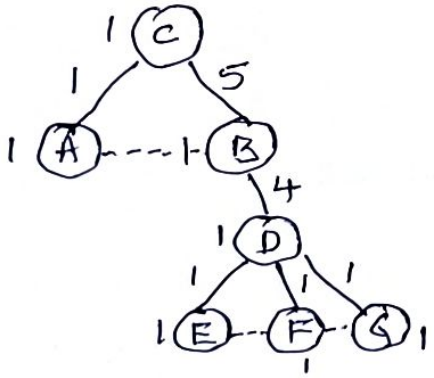

## For A node



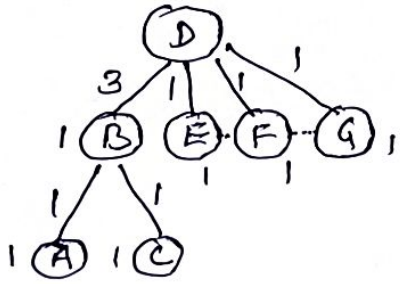

## For Node B.



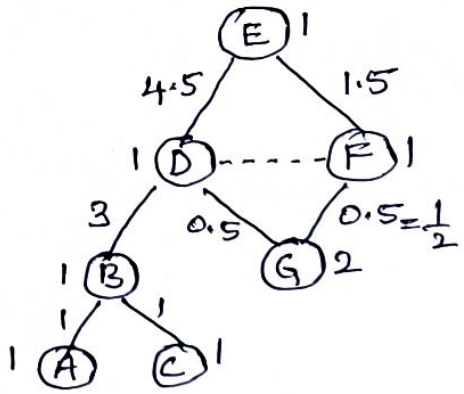

## For Node C



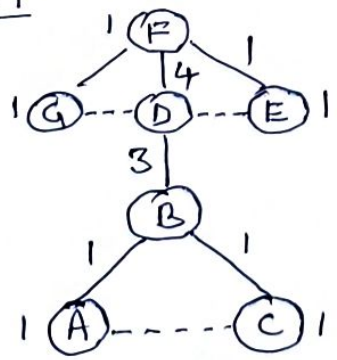

## For Node D



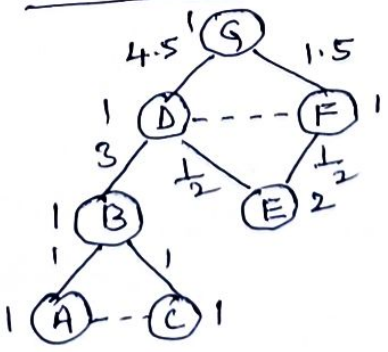

## Node E



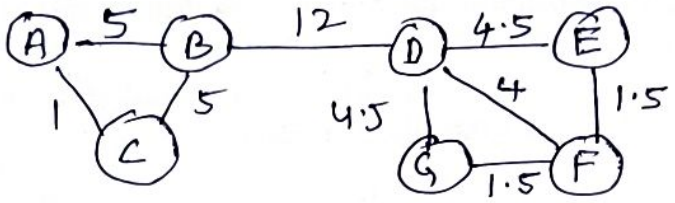

## Node F




## For Node G

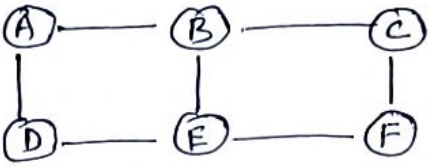


| Edges | Betweenness centralities | | | | | | | Total | Total/2 |
|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | E | F | G | | |
| AB | 5 | 1 | 0 | 1 | 1 | 1 | 1 | 10 | 5 |
| AC | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 2 | 1 |
| BC | 0 | 1 | 5 | 1 | 1 | 1 | 1 | 10 | 5 |
| BD | 4 | 4 | 4 | 3 | 3 | 3 | 3 | 24 | 12 |
| DG | 1 | 1 | 1 | 1 | 0.5 | 0 | 4.5 | 9 | 4.5 |
| DF | 1 | 1 | 1 | 1 | 0 | 4 | 0 | 8 | 4 |
| DE | 1 | 1 | 1 | 1 | 4.5 | 0 | 0.5 | 9 | 4.5 |
| EF | 0 | 0 | 0 | 0 | 1.5 | 1 | 0.5 | 3 | 1.5 |
| FG | 0 | 0 | 0 | 0 | 0.5 | 1 | 1.5 | 3 | 1.5 |

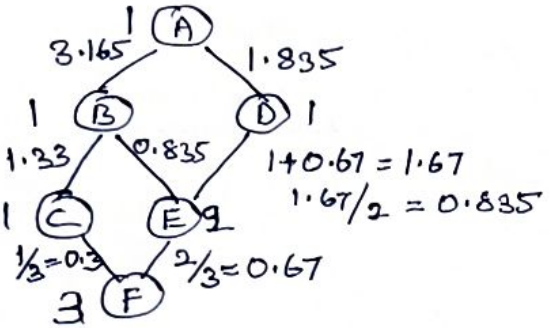



Remove BD – Max. betweenness :, Communities {A, B, C}
{D, E, F, G,}

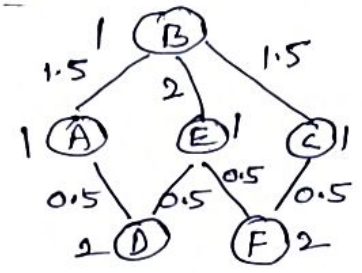
Example: Calculate betweenness centrality for the edges solve using Girvan Newman Algorithm.



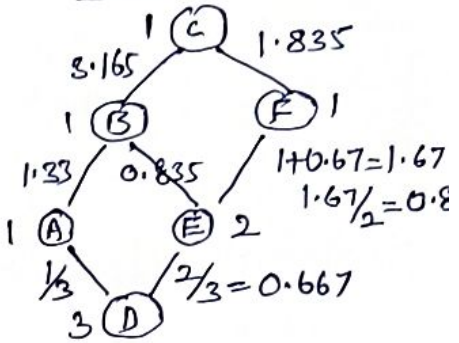

Node 'A'



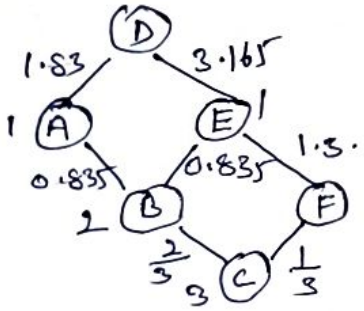

Node: B.



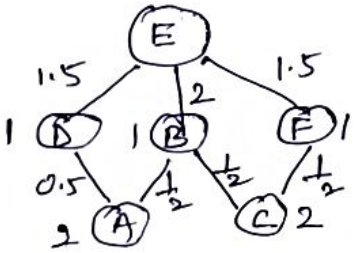

Node C



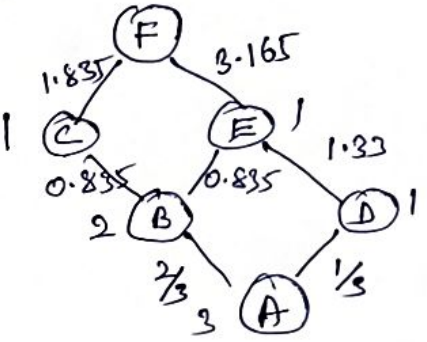

Node: D



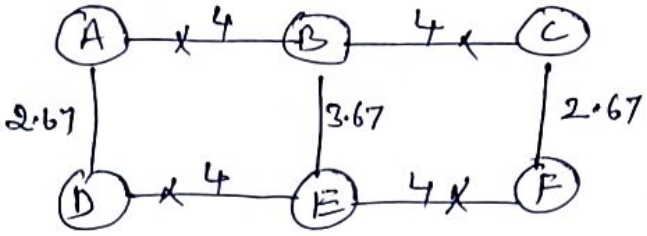

Node E



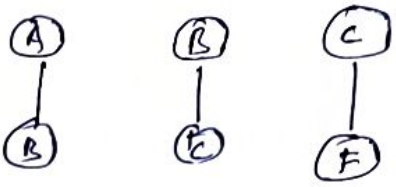

Node: F




| Edges | Edge betweenness | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | E | F | Total | Total/2 |
| AB | 3.165 | 1.5 | 1.33 | 0.835 | 0.5 | 0.066 | 8 | |
| AD | | | | | | | 5.33 | |
| BC | | | | | | | 8 | |
| BE | | | | | | | 7.34 | |
| CF | | | | | | | 5.33 | |
| DE | | | | | | | 8 | |
| EF | | | | | | | 8 | |




There are three communities




## Exercices:

(1) problem.

